Status	Finished
Started	Tuesday, 20 October 2026, 8:46 PM
Completed	Tuesday, 20 October 2026, 8:49 PM
Duration	2 mins 52 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Binary search algorithm is not efficient when the data elements more than 1500 ✓

b. Must use a sorted array

c. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

d. There must be a mechanism to access middle element directly

Question **2**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)

b. O(n log n)

c. O🥱

d. O(log n) ✓

Question **3**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n-1

b. n ✓

c. n/2

d. log n

Question **4**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the list contains duplicate elements
- b. When the search algorithm finishes
- c. When the list is sorted
- d. When the element is found in the list ✓

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is unsorted
- b. When the list is sorted ✓
- c. When the list is very small
- d. When the list contains duplicate elements

Question **6**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search has a lower time complexity on large, sorted lists ✓
- b. Binary search does not require dividing the list
- c. Binary search can find multiple instances of the target element
- d. Binary search works on unsorted lists

Question **7**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O👎
- b. O(n log n)
- c. O(log n)
- d. O(1) ✓

Question **8**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

- a. The search ends successfully ✓
- b. The search continues in the left sublist
- c. The search continues in the right sublist
- d. The list is sorted

Question **9**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Linear search
- b. Binary search ✓
- c. Sequential search
- d. Hash search

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Binary search
- b. Linear search ✓
- c. Hashing
- d. Dynamic search

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It is slower than linear search for small lists
- b. It does not work with negative numbers
- c. It requires the list to be sorted ✓
- d. It can only be applied to large lists

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Quick search
- b. Linear search ✓
- c. Merge search
- d. Bubble search

Question **13**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

- a. Discovering
- b. Searching ✓
- c. Finding
- d. Mining

Question **14**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By comparing each element in the list sequentially ✓
- b. By using a hash function
- c. By dividing the list into halves
- d. By sorting the list first

Question **15**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

- a. 94 and 99
- b. 90 and 99 ✓
- c. 89 and 94
- d. 90 and 100